Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2023

Course: B.Tech
Code: 100302

Subject: Analog Electronic Circuits

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Answer the following questions (any seven only):*** **[2 x 7 = 14]**

(a) Draw the pn junction diode VI characteristics.
(b) Mention the advantages of Full Wave Rectifier.
(c) What are the different configurations of BJT?
(d) What is stability factor?
(e) What is MOSFET? Classify the types of MOSFET.
(f) What is meant by pinchoff voltage?
(g) Define an operational amplifier.
(h) Define Slew rate.
(i) Mention some of the non-linear applications of op – amp.
(j) List the features of instrumentation amplifier.

Q.2 (a) What is a PN Junction? Explain the formation of depletion layer in a PN junction. **[7]**
(b) Draw the circuit diagram of half wave rectifier and explain its operation with the help of waveforms. **[7]**

Q.3 (a) If the base current in a transistor is 20μA when the emitter current is 6.4mA, what are the values of α and β? Also calculate the collector current. **[7]**
(b) Why hybrid model is used for the analysis of BJT amplifier at low frequencies? Draw the hybrid model for CE transistor and derive the parameters. **[7]**

Q.4 (a) Draw and explain the small signal model of FET at low frequency. **[7]**
(b) With the help of neat diagram, explain the operation and characteristics of n-channel enhancement type MOSFET. **[7]**

Q.5 (a) What are the factors that affect the frequency stability of an oscillator? How frequency stability can be improved in oscillators. **[7]**
(b) Compare different configurations of differential amplifier. **[7]**

Q.6 (a) Write the design steps of the first order low pass filter and draw its circuit. **[7]**
(b) With the help of neat and clear sketch, explain the working of a MOS transistor as a switch. **[7]**

Q.7 Explain the basic building block diagram of an op-amp. Write short note on input bias current, output bias current, slew rate and output offset voltage of an op-amp. **[14]**

Q.8 (a) Explain the working of series clipping circuit to clip the input sinusoidal signal above reference level. Draw the waveforms and transfer characteristics. **[7]**
(b) Compare BJT and FET on different parameters. **[7]**

Q.9 (a) Write Short notes on (i) Zero Crossing Detector (ii) Precision Rectifier **[7]**
(b) Explain the function of Square-wave and triangular –wave generators. **[7]**

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